Is there Termhood Beyond Memorisation? A Feature-Based Approach with Term-Exclusion Cross-Validation

¿Hay terminologicidad más allá de la memorización? Un enfoque basado en rasgos con exclusión de términos y validación cruzada

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Abstract: Automatic Term Extraction (ATE) is generally framed as a supervised sequence-labelling task, in which models learn to identify terminological units from annotated examples. The rise of neural architectures has driven significant advancements, often relegating the role of linguistic features to a secondary position, despite their being essential for describing the nature and behavior of terminological units. This work presents a model based on Conditional Random Fields (CRF) enriched with linguistic and statistical features and designed to model the terminological unit beyond superficial lexical forms. To distinguish between memorization and genuine terminological competence, we adopt a strict term-exclusion protocol. Using sentence-level cross-validation, we ensure that each test partition contains unseen terms, allowing us to evaluate the model's capacity to generalize to new units. The results demonstrate that, under this strict evaluation, the integration of linguistic features is fundamental, achieving a Macro F1 of ~ 0.79. Furthermore, inspecting the CRF weights allows us to interpret the model's terminological competence, explicitly identifying which linguistic and statistical features condition or determine termhood. Collectively, this proposal offers a transparent and replicable methodology for ATE, focused on interpretability and generalization over unseen lexicon.
Keywords: Automatic Term Extraction, Terminology, Termhood, Machine Learning

Resumen: La extracción automática de términos (EAT) se formula, generalmente, como una tarea de etiquetado de secuencias supervisada en la que los modelos aprenden a identificar unidades terminológicas a partir de ejemplos anotados. El auge de las arquitecturas neuronales ha impulsado avances significativos, a menudo relegando a un segundo plano el papel de los rasgos lingüísticos, esenciales para describir la naturaleza y el comportamiento de las unidades terminológicas. Este trabajo presenta un modelo basado en Conditional Random Fields (CRF) enriquecido con rasgos lingüísticos y estadísticos y diseñado para modelar la unidad terminológica más allá de formas léxicas superficiales. Para distinguir entre memorización y competencia terminológica real, adoptamos un estricto protocolo de exclusión léxica. Mediante validación cruzada a nivel de oración, garantizamos que cada partición de prueba contenga términos no vistos, permitiendo evaluar la capacidad del modelo para generalizar a nuevas unidades. Los resultados demuestran que, bajo esta evaluación estricta, la integración de rasgos lingüísticos es fundamental, logrando una Macro F1 de ~ 0.79. Además, la inspección de los pesos del CRF nos permite interpretar la competencia terminológica del modelo, identificando explícitamente qué rasgos lingüísticos y estadísticos condicionan o determinan la terminologicidad. En conjunto, esta propuesta ofrece una metodología transparente y replicable para la EAT, enfocada en la interpretabilidad y la generalización sobre léxico no visto.
Palabras clave: Extracción automática de términos, Terminología, Terminologicidad, Aprendizaje automático

1 Introduction

Automatic Term Extraction (ATE) is a central task in domain–specific Natural Language Processing (Jacquemin and Bourigault, 2005; Astrakhantsev, Fedorenko, and Turdakov, 2015; Heylen and De Hertog, 2015) aimed at identifying lexical units that encode specialised concepts and support the organisation and transmission of expert knowledge (Cabré, 1999; Estopà, 1999; Stanković et al., 2016). Over the past decades, ATE has evolved from early statistical approaches based on frequency and association measures to linguistically driven rule-based methods, and more recently to supervised sequence-labelling architectures, increasingly implemented through neural models (Astrakhantsev, Fedorenko, and Turdakov, 2015; Lossio-Ventura et al., 2016; Tran et al., 2023; Blandón Andrade et al., 2025).

As these approaches increasingly shape ATE research, evaluation becomes a cornerstone of methodological design (Bernier-Colborne, 2012; Terryn et al., 2020). However, assessing performance in ATE is not merely a matter of measuring predictive accuracy: it must reflect how terminology behaves in text. Thus, an evaluation protocol that ignores this risks rewarding systems for recovering familiar lexical items rather than identifying the linguistic properties that distinguish terms from general vocabulary. This issue has already been observed in related NLP tasks where models achieve deceptively high performance by exploiting superficial lexical overlap instead of learning abstract linguistic relations (Levy et al., 2015; Babl et al., 2025). ATE is especially vulnerable to this issue because terminological lexicons are highly uneven and relatively limited in variety in contrast to the rest of lexical units, i.e., once a term enters specialised discourse, it tends to recur frequently, which ultimately makes it difficult to distinguish genuine terminological competence from lexical familiarity.

This paper examines whether a supervised ATE model can infer *termhood* from linguistic and quantitative evidence in specialised discourse rather than relying on prior exposure to particular lexical forms. To address this question, we adopt an evaluation protocol that removes any lexical overlap between training and test sets in order to assess whether the model can identify terms it has never encountered during training.

RQ: Do supervised ATE models genuinely generalise to unseen terms, or do conventional evaluation practices simply reward lexical memorisation?

From this question we derive two hypotheses:

1. Performance in conventional ATE settings is overestimated due to lexical overlap between training and test partitions.
2. When such overlap is prevented, a model that exploits linguistic, contextual and distributional cues associated with termhood should maintain stable performance.

Based on these hypotheses, our study pursues three objectives:

1. Evaluate a Conditional Random Field (CRF) model for ATE under controlled cross-validation scenarios.
2. Implement a term-exclusion protocol ensuring strict separation of term types between training and test sets.
3. Establish a reproducible evaluation framework that distinguishes memorisation from genuine terminological inference.

2 Related work

2.1 Basic concepts

Automatic Term Extraction (ATE)—also referred to as *Automatic Term Recognition* (Acosta, Aguilar, and Infante, 2015; Astrakhantsev, Fedorenko, and Turdakov, 2015) or *Terminology Mining* (Daille, 2003)—is a computational task concerned with identifying lexical units that encode specialised concepts within domain-specific discourse (Zhang, Lu, and Sui, 2007; Zhang, Song, and Fang, 2010; Frantzi, Ananiadou, and Mima, 2000; Heylen and De Hertog, 2015). ATE systems have been developed for a wide range of domains, including highly-regulated and prolific ones such as Biomedicine (Lossio-Ventura et al., 2016) or Law (Martín Chozas and Calleja Ibáñez, 2018), and have traditionally been motivated by the need to reduce the manual effort and potential bias involved in terminological

work (Heylen and De Hertog, 2015). As such, ATE constitutes a principal entry point in the terminology workflow, supporting the development of terminological resources (Acosta, Aguilar, and Infante, 2015)—from glossaries and terminological databases to taxonomies and ontologies (da Silva Conrado et al., 2014)—and enabling downstream applications such as specialised translation and the automatic indexing of domain-specific documents (Casas Gómez, 2022). Yet ATE inherits a foundational problem from Terminology: there is no consensus on the nature of the linguistic object it seeks to extract. Any computational system designed to identify terms must, implicitly or explicitly, answer questions such as:

- what constitutes a *term* as opposed to other lexical items;
- how such units can be recognised—manually or automatically—in real corpora; and
- which linguistic, distributional or cognitive properties enable their differentiation.

These questions persist because, although the judgement of whether a lexical unit functions as a term is ultimately pragmatic and context-dependent—at least from a linguistically oriented perspective, as opposed to the foundational settings of Terminology that conceived terms as stable, predefined units (Cabré, 1999)—terms do not systematically exhibit overt formal markers that disclose their specialised status. Nonetheless, their use in specialised discourse leaves identifiable traces—morphosyntactic regularities, distributional profiles and contextual patterns—that signal their epistemic role (L'Homme, 2020). To operationalise this distinction, Computational Terminology introduces two complementary constructs (Kageura and Umino, 1996):

1. *unithood*, or the degree to which a sequence forms a stable linguistic unit, and
2. *termhood*, or the extent to which that unit is conceptually salient within a specialised domain.

While the first captures syntagmatic cohesion—e.g. noun-headedness and modifier–head configurations—the second reflects domain relevance and epistemic centrality (Nazar, 2016). These constructs underpin the dominant computational formulation of ATE as a two-phase pipeline (Vivaldi and Rodríguez, 2011; Ananiadou and McNaught, 2005; Kockaert et al., 2015): (i) candidate extraction, which retrieves potential units using linguistic templates or n-gram heuristics, and (ii) candidate validation, which determines which candidates qualify as terms via statistical evidence or supervised learning (Tran et al., 2024). More recent neural approaches learn these distinctions implicitly, but often conflate terminological inference with lexical memorisation, leaving unresolved the question of whether models can infer what makes a lexical unit a term rather than merely recognise previously seen forms.

2.2 Methods

Research in ATE has progressed through successive methodological paradigms (da Silva Conrado et al., 2014; Lossio-Ventura et al., 2016; Tran et al., 2023), distinguished by the type of supervision involved and by the linguistic and/or statistical properties leveraged for identifying specialised vocabulary. Early statistical approaches relied on frequency-based metrics and association measures to rank candidate terms, including TF–IDF, Weirdness and other contrastive indices (Astrakhantsev, Fedorenko, and Turdakov, 2015). Although effective in identifying salient lexical units, these approaches struggled with multiword expressions and domain-induced semantic shifts, often favouring high recall at the expense of precision. In order to mitigate these limitations, rule-based and linguistic methods incorporated morphosyntactic constraints (e.g., *adjective+noun* or *noun+preposition+noun* templates) supported by PoS tagging, chunking and lemmatisation (Tran et al., 2023; Koza Orellana, 2015). Although these achieve high precision in controlled settings, they require substantial expert knowledge and show limited portability across domains and languages.

Machine-learning approaches, on the other hand, frame ATE as a task in which models learn lexical, syntactic and distributional patterns associated with *termhood* from annotated data. Early work explored supervised classifiers such as SVMs (Bol-

shakova, Loukachevitch, and Nokel, 2013), while unsupervised approaches used clustering algorithms (e.g. k-means) to group candidate terms based on distributional similarities (Zhang, Lu, and Sui, 2007; Lossio-Ventura et al., 2016). Under this paradigm, Conditional Random Fields (CRFs) emerged as a natural fit for modelling term boundaries, allowing each token to be classified as part of a term or not while maintaining sequence-level consistency. Their application spans foundational work on segmentation and terminology (Peng, Feng, and McCallum, 2004) and subsequent extensions to term recognition across domains (Zhang, Song, and Fang, 2010; Fkih, Omri, and Toumia, 2012). Supervised ATE models typically depend on hand-crafted feature engineering, where term candidates are represented through a rich set of linguistic and statistical descriptors. Common linguistic features include PoS tags, lemmatisation, token shape, orthographic markers, word length, and constituent structure, capturing regularities such as noun-headedness and modifier–head configurations (Koza Orellana, 2015).

Despite the rise of neural architectures and Transformer-based embeddings, which infer contextual semantic properties without explicit feature design, CRFs remain relevant for ATE due to their interpretability and their suitability for low-resource conditions. CRFs make explicit how termhood emerges from observable linguistic evidence—a distinction central to this study (Zheng, Zhao, and Yang, 2009). Rather than relying on massive pretraining or implicit representations, the present work employs a CRF enriched with morphosyntactic, orthographic, distributional and contextual features, enabling a controlled examination of whether such signals are sufficient for modelling terminological competence. Feature-based CRFs therefore constitute an appropriate testbed for isolating the contribution of linguistic and quantitative cues to the inference of *termhood*. This motivates the methodological choice adopted here: evaluating a single CRF architecture under a controlled term-exclusion protocol to determine whether a model can infer *termhood* for previously unseen lexical forms.

3 Methodology

This section details the methodological design adopted to assess ATE as a supervised sequence-labelling task. Our approach is grounded in two complementary principles. First, we model ATE following a three-class `BIO` labelling scheme (§ 3.1). Second, the experimental setup is applied to a domain-specific corpus (§ 3.2), which constitutes the basis for feature-selection, training (§ 3.3) and evaluation (§ 3.4).

3.1 Task

ATE can be conceptualised as a supervised classification task applied at the token level, in which each lexical item receives a label indicating its participation in a domain-specific term. This classification can be operationalised in two ways: categorically, through discrete decisions that assert whether a token belongs to a term, or along a *continuum* that assigns degrees of terminicity (Nazar, 2016). In this experiment, ATE is implemented as a supervised sequence labelling problem in which each token in a sentence receives a label marking whether it participates in a domain-specific term. Each sentence in the corpus is represented as a sequence of tokens paired with BIO labels. Multiword terms are annotated by marking the first token as (`B-T`) (beginning of term) and subsequent tokens as (`I-T`) (inside term), while all tokens that do not belong to a term are labelled as outside of the term (`O`).

This scheme allows for the preservation of the internal structure and boundaries of both single- and multiword terms, while still understanding ATE as a sequence-labelling task. The resulting representation provides an explicit training signal that allows the supervised model to observe the linguistic environments in which terms occurs. Under this formulation, the core problem is to decide, for each token in context, whether it initiates, continues or does not participate in a terminological unit within specialised discourse.

3.2 Dataset

The dataset used in this study consists of a domain-specific corpus in Corpus Linguistics, compiled with the aim of capturing the conceptual and terminological repertoire characteristic of this field. The corpus includes 50 full-text scientific texts published in specialised venues, selected according to three

criteria: (i) thematic relevance, ensuring that the texts discuss core topics such as corpus design, annotation, representativeness, quantitative methods and linguistic description from a corpus-based perspective; (ii) disciplinary homogeneity, restricting the corpus to texts authored within the field to maintain conceptual coherence; and (iii) textual completeness, privileging full articles over abstracts or sections to preserve terminological context. Only publications from 2015 onwards were included, ensuring that the corpus reflects contemporary practices and the current terminological landscape of the discipline (Kÿto, Rudanko, and Smitterberg, 2000; Durán Muñoz, 2011).

A distinctive property of terminological datasets is not merely the rarity of terminological tokens but also the restricted diversity of the terminological lexicon. In specialised discourse, a relatively small set of terms tends to recur frequently, reflecting their centrality to the domain; however, the number of distinct term types remains limited (Corpas Pastor and Seghiri Domínguez, 2010). This lexical concentration is quantified in Table 1: despite covering 1.4 million tokens, the corpus contains only 467 distinct normalized term types. More importantly, the term density—the proportion of tokens that are term occurrences—is 3.72% (reconstructed from `BIO` tags) or 4.16% (at token-level), meaning that terminology is lexically constrained, reducing the opportunities for models to encounter new forms.

Measure	**Token-level**	**Reconstructed**
No. texts	50	50
No. sentences	54,060	54,060
No. tokens	1,397,937	1,397,937
Term tokens	58,214	52,023
Term types	–	467
Term density	**4.16%**	**3.72%**

Table 1: Descriptive statistics of the annotated corpus.

Figure 1 displays the terminological saturation curve derived from token-level processing of the entire corpus (Corpas Pastor and Seghiri Domínguez, 2010). The curve plots the cumulative number of unique term types as a function of the number of tokens observed, enabling the identification of key inflection points where the incorporation of new terminology decreases sharply. Notably, the plot reveals the 80% saturation threshold, beyond which the lexical growth rate slows significantly and the corpus enters a plateau region. This behaviour confirms that most domain-specific terminology appears relatively early in the dataset, with diminishing returns in later portions of the corpus.

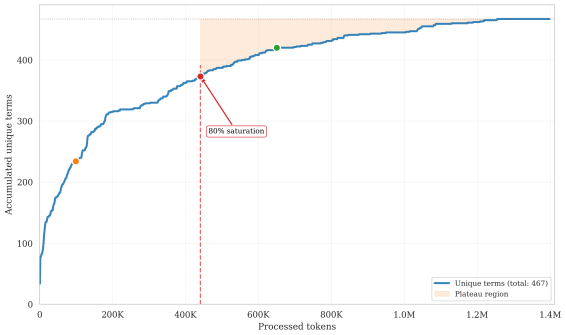


Figure 1: Terminological saturation curve computed from the corpus.

As shown in Table 2, single-word terms (SWTs) constitute the majority of both the vocabulary (59.5%) and of actual usage (89.9% of term occurrences). Multi-word terms (MWTs), though numerous in types (189 distinct forms), contribute only 10.1% of term tokens. This distribution follows a strong Zipfian pattern: a small core of frequent single-word terms (e.g., domain-specific nouns like *corpus*, *annotation*) accounts for most terminological density, while a longer tail of multi-word expressions appears sporadically. The average term length of 1.12 tokens further underscores the predominance of monolexical units, i.e., SWTs.

	Types	**Tokens**
SWTs	278 (59.5%)	46,769 (89.9%)
MWTs	189 (40.5%)	5,254 (10.1%)
Avg. term length	1.12 tokens	
Max. term length	7 tokens	

Table 2: Distributional characteristics of normalized terms in the dataset.

The corpus was annotated using *Doccano*, an open-source platform designed for annotation purposes which allows for sequence-

labelling tasks[1]. Each token was tagged under a single class, *TERM*, enabling a fine-grained, token-level perspective on terminological structure. This annotation setup captures both SWTs and MWTs, ensuring that terminological boundaries are modelled directly rather than inferred from surface n-gram heuristics. Once the manual annotation was completed, the labelled data were automatically converted into the `BIO` scheme, preserving distinguishing term-initial positions (`B-T`) from internal ones (`I-T`). The annotation was performed by a single expert annotator with specialised training in Terminology and Corpus Linguistics.

3.3 Model

Conditional Random Fields (CRFs) are a class of discriminative probabilistic graphical models designed for structured prediction tasks, particularly sequence labeling (Lafferty, McCallum, and Pereira, 2001). In this work, we employ a linear-chain CRF to infer the terminological status of lexical units within scientific texts.

The model leverages explicitly encoded linguistic information through a set of features, where each token in a sequence encapsulates morphological, syntactic, and distributional properties theoretically correlated with *unitehood* and *termhood*. The feature set comprises 40 effective descriptors (excluding identifiers and the target label), as detailed in Table 3.

Category	**Proportion**
Morphological	45.00%
Syntactic	12.50%
Distributional	20.00%
Contextual	15.00%
Lexical	7.50%
Total	**100%**

Table 3: Distribution of the features used in the CRF model.

These categories are linked to key properties of terminological recognition:

- **Morphosyntactic Cues for Unithood:** Captures stable multi-word patterns via POS tags, dependency relations, and morphosyntactic descriptors (e.g., `pos`, `dep`, `morph`). Orthographic regularities are signaled as well (e.g., `has_hyphen`, `shape`, and `is_title`).
- **Lexical and Formal Signals:** Models unit stability through basic token properties (e.g., `lower`, `token_len`, `is_upper`) and affixal information (`prefix_3`, `suffix_3`).
- **Statistical Indicators for Termhood:** Operationalises domain salience using distributional metrics (e.g., `freq_global`, TF–IDF, dispersion) and terminology-specific measures (e.g., `cvalue`).
- **Contextual and Semantic Evidence:** Identifies local cohesion through immediate context (e.g., `prev_token`, `next_pos`) and provides semantic guidance via entity tags (e.g., `ent_type`) and pattern flags (e.g., `is_in_definition_pattern`).
- **Syntactic Configuration Features:** Supplies evidence on a token's role within a potential term using relational features (e.g., `is_in_noun_phrase`, `has_compound_modifier`).

In line with prior terminology research (Kageura and Umino, 1996), our feature design reflects the distinction between the structural cohesion of a candidate unit (i.e., *unithood*) and its domain-oriented salience (*termhood*). The feature set operationalises these two dimensions through complementary evidence: morphosyntactic and formal indicators (e.g., PoS configurations, dependency-based relations, modifier–head structure, token shape and capitalisation) capture the internal stability of lexical sequences, whereas statistical, contextual and semantic cues (including domain-sensitive frequency patterns, TF–IDF, C-Value, definitional-context signals and precomputed semantic labels) target the distributional behaviour associated with specialised lexical status. Taken together, these features allow the model to integrate structural regularities with domain-sensitive signals in a unified sequence-labelling framework.

3.4 Evaluation

Given the lexical constraints inherent in terminological datasets (§ 3.2), standard ran-

[1]https://doccano.web.webis.de/

dom splits allows for an evaluation scenario in which models repeatedly encounter the same terminology during training and testing. To mitigate this problem, we adopt a term-exclusion or split protocol that ensures lexical separation between training and test data. Rather than randomly splitting at the sentence level, we implement a strict lexical cross-validation procedure that operates on complete term types.

The evaluation protocol is designed to:

- Ensure no overlapping of complete terms between training and test sets within each fold, eliminating direct lexical memorisation;
- Test models' ability to identify unseen term types based on linguistic and quantitative patterns rather than token-level familiarity;
- Maintain natural sentence structure while preserving the distribution of terminological phenomena.

This approach separates recognition from generalisation: high performance under random splits may reflect familiarity with specific lexical tokens, whereas sustained performance under term-exclusion indicates the ability to identify terms based on feature-based evidence. By enforcing this constraint, the evaluation provides a principled test of whether models internalise terminological properties (e.g., morphological, distributional, or syntactic patterns) rather than reproducing memorised lexical mappings. This is formalised in Algorithm 1, which implements a 5-fold cross-validation scheme with strict lexical separation:

Algorithm 1: Cross-validation protocol based on term-level exclusion

1. **Input:** Annotated corpus C and number of folds K.
2. **Term extraction:** Retrieve the set of multiword terms T from the `BIO` labels.
3. **Term–sentence mapping:** For each term $t \in T$, collect all sentences in which t occurs.
4. **Lexical partitioning:** Randomly divide T into K disjoint subsets $T_1, \ldots, T_K$.
5. **Fold construction:** For each $k \in \{1, \ldots, K\}$:
   - Define $Test_k$ as the set of sentences containing any term in T_k.
   - Set $Train_k = C \setminus Test_k$.
   - Remove `B-T`/`I-T` tags in $Train_k$ for tokens belonging to terms in T_k.
6. **Exclusion check:** Verify that no term from T_k remains annotated in $Train_k$.
7. **Output:** A set of K independent train–test splits.

4 Results

The CRF exhibits high precision for the `B-T` class (0.96 ± 0.04) and satisfactory recall (0.78 ± 0.11), indicating that the model reliably identifies the onset of terms when contextual and morphosyntactic cues are sufficiently informative. In contrast, while recall for the `I-T` class remains strong (0.70 ± 0.23), precision drops markedly (0.42 ± 0.19), suggesting that once a term has been initiated, the model frequently overextends its boundaries. This behaviour reflects a systematic difficulty in delimiting the full span of multiword expressions, even when their internal cohesion is correctly captured. The overall macro F1 score of 0.79 ± 0.10 confirms robust performance under the strict lexical-split regime, though the relatively high variance across folds reveals heterogeneous term distributions and domain-specific idiosyncrasies in the dataset.

Label	P	R	F1
`B-T`	0.96 ± 0.04	0.78 ± 0.11	$\mathbf{0.86 \pm 0.08}$
`I-T`	0.42 ± 0.19	0.70 ± 0.23	$\mathbf{0.52 \pm 0.21}$
`O`	1.00 ± 0.00	1.00 ± 0.00	$\mathbf{1.00 \pm 0.00}$
Macro F1		$\mathbf{0.79 \pm 0.10}$	

Table 4: Aggregated results of the cross-validation evaluation (mean ± standard deviation) across folds.

The ablation study further substantiates these findings. Removing distributional features reduces F1 by 28.0%, while excluding morphological or syntactic information degrades performance by 17.2% and 9.2%, respectively. These drops are especially pronounced in the detection of internal boundaries of multiword terms, confirming that

Group	F1	ΔF1 (%)
All	0.50 ± 0.03	–
- Context	0.50 ± 0.02	-0.6
- Distributional	$\mathbf{0.36 \pm 0.04}$	$\mathbf{+28.0}$
- Lexical	0.48 ± 0.04	$+3.4$
- Morphology	0.41 ± 0.01	$+17.2$
- Syntax	0.46 ± 0.09	$+9.2$

Table 5: Ablation study: removing distributional features *degrades* F1 by 28.0%; morphology removal *degrades* it by 17.2%.

such features are essential for modelling unithood. In contrast, ablating surface-level lexical cues yields only marginal differences, showing that the model does not compensate through orthographic heuristics or lexical memorisation. Overall, the ablation analysis reinforces the central claim of this work: the CRF derives its performance from the interaction of linguistic and domain-sensitive statistical features, providing a principled basis for hybrid neural–CRF architectures in future research.

5 Discussion

The CRF-based ATE model achieved a macro F1 of approximately 0.79 under a rigorous evaluation setting. This indicates a modest overall ability to separate terms from other lexical units in text, despite the challenging constraints imposed by the evaluation protocol. To interpret this performance in detail, we consider the class-specific results for the terminological labels (`B/I-T`). In practical terms, the system is conservative in initiating terms – it rarely flags a token as a term beginning unless there is strong evidence, leading to few false term starts (hence the high precision). This caution is beneficial because a spurious term start would yield an incorrect term extraction. On the other hand, the model shows high recall for I-T labels, suggesting that once a term has begun, the model is adept at capturing the remaining tokens of the term. In other words, it tends to label interior tokens of multi-word terms correctly, seldom leaving out tokens that should be part of a term. This implies the CRF has learned strong sequential patterns for continuing terms, likely leveraging context to decide that a token belongs inside an existing term. The trade-off is that precision for I-T might be lower than its recall – the model sometimes labels a token as inside a term when it is not, potentially over-extending a term boundary. Such behavior aligns with the known importance of boundary detection ATE, where even advanced language models struggle with precise term span identification (Chun et al., 2025)).

A key aspect of our evaluation was a strict term-exclusion cross-validation protocol, wherein no term is present in both the training and test sets. This means the model was never evaluated on a term it had seen during training, eliminating any chance to simply memorize term appearances. Such a protocol provides a stringent test of generalization. By prohibiting lexical overlap, we ensure that the model's performance reflects its ability to recognize novel terms based on learned patterns, rather than recalling known terms from training data as a result of lexical overlapping (Chun et al., 2025). The relevance of this protocol becomes evident when contextualised within the broader literature. As has been repeatedly noted, despite shared reliance on precision, recall and F1-scores, ATE evaluation lacks methodological standardisation (Bernier-Colborne, 2012; Rigouts Terryn, Hoste, and Lefever, 2022). Studies differ in the construction of their gold standards, the minimum and maximum lengths of allowed terms, the treatment of MWT sequences, and whether partial matches are accepted as correct or not. These inconsistencies preclude direct comparison of reported scores and have contributed to a fragmented view of system performance in the field (Terryn et al., 2020; Rigouts Terryn, Hoste, and Lefever, 2022).

The suggested model is consistent with other supervised approaches (Zheng, Zhao, and Yang, 2009; Zhang, Song, and Fang, 2010). It's worth noting that some recent studies using deep neural models (e.g., BERT fine-tuning or BiLSTM-CRF with embeddings) have pushed performance slightly higher on certain benchmarks (Zhang, Zhou, and Li, 2020; Chun et al., 2025). However, those gains often come at the cost of requiring much larger training data and the risk of losing some interpretability. Adittionally, our work is aligned with research highlighting the continued relevance of explicit linguistic knowledge in ATE systems (Tran et al., 2023). We incorporate many of these cues within a learning framework and complement them with corpus-driven statistical indica-

tors. However, the language-agnostic nature of several of our features (e.g., frequency-based measures and general POS categories) raises the question of how the approach might scale beyond English. Some cues—such as suffixal patterns and prototypical multi-word term structures—are language-specific, which suggests that adaptations will be required for morphologically richer or analytically different languages. The broader literature acknowledges that multilingual supervised ATE remains constrained by the scarcity of domain-diverse, high-quality terminological corpora (Bernier-Colborne, 2012; Rigouts Terryn, Hoste, and Lefever, 2022). Existing resources, such as the ACTER dataset (Terryn et al., 2020), constitute an important step forward, but coverage is still limited in terms of both languages and specialised domains. This bottleneck partly explains why most supervised ATE research focuses on English and why generalisation across domains still remains an unresolved challenge.

6 Conclusion and Future Work

This study has demonstrated that ATE is a task which can be modelled as a principled inference task rather than a memorisation exercise, provided that evaluation protocols enforce true lexical independence between training and testing data. The behaviour of the system offers a nuanced view of terminological structure. Its conservative approach to detecting term onsets leads to high precision for initial term labels, while its strong sequential modelling capabilities ensure high recall for interior components of MWT, which highlights the long-established complexity of boundary delimitation in ATE: although the internal cohesion of terms is well captured, determining their exact span remains a challenging aspect of the task, which is ultimately intrinsic to the nature of terminological units (§ 3.1).

Moreover, the characteristics of terminological corpora themselves contribute to this difficulty. As discussed in the dataset analysis, specialised corpora are highly repetitive and saturated with recurrent lexical patterns and morphological variants (§ 3.2). The fact that the model achieves competitive results under strict lexical separation reinforces the conclusion that its performance is not merely an artefact of corpus repetition, but evidence of genuine generalisation over systematic encoded features of termhood.

Beyond performance, the results aim to contribute methodological clarity to a field often characterised by heterogeneous evaluation practices (Bernier-Colborne, 2012; Terryn et al., 2020; Rigouts Terryn, Hoste, and Lefever, 2022). Overall, the findings invite a reconsideration of how success in ATE is measured and achieved. The evidence presented here indicates that models capable of leveraging linguistic cues can extract novel terminology reliably, provided that evaluation frameworks are aligned with that objective. In this sense, the contribution lies not only in the performance of the model itself, but in suggesting conditions under which such performance can be meaningfully interpreted and compared.

Future work will explore how these insights may be extended through hybrid architectures that combine the interpretability of feature-based models with the representational capacity of neural contextual embeddings (Zhang, Zhou, and Li, 2020). Such integration may alleviate boundary-detection errors by capturing semantic regularities that exceed the scope of handcrafted features alone. Additionally, further research will examine alternative evaluation scenarios, including fuzzy similarity-based protocols that account for partial or near-equivalent term matches. These scenarios may better reflect the gradience inherent in terminological variation, offering a more nuanced view of system performance beyond strict span identity.

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[Anonymised]

References

Acosta, O. L., C. A. Aguilar, and T. Infante. 2015. Reconocimiento de términos en español mediante la aplicación de un enfoque de comparación entre corpus. *Linguamática*, 7(2):19–34.

Ananiadou, S. and J. McNaught. 2005. *Text mining for biology and biomedicine*. Artech.

Astrakhantsev, N. A., D. G. Fedorenko, and D. Y. Turdakov. 2015. Methods for automatic term recognition in domain-specific text collections: A survey. *Programming and Computer Software*, 41(6):336–349.

Babl, F., M. Hennen, J. Murauer, and M. Geierhos. 2025. Random splitting negatively impacts ner evaluation: Quantifying and eliminating the overestimation of ner performance. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 9724–9738.

Bernier-Colborne, G. 2012. Defining a gold standard for the evaluation of term extractors. In *Proceedings of the colabTKR workshop Teminology and Knowledge Representation, Istanbul, Turkey.*

Blandón Andrade, J. C., C. M. Medina Otálvaro, C. M. Zapata Jaramillo, and A. Morales Ríos. 2025. Approaches, tools, algorithms, and methods for automatic term extraction: A systematic literature mapping. *Journal of Intelligent & Fuzzy Systems*, page 18758967251392652.

Bolshakova, E., N. Loukachevitch, and M. Nokel. 2013. Topic models can improve domain term extraction. In *European Conference on Information Retrieval*, pages 684–687. Springer.

Cabré, M. T. 1999. *Terminology: theory, methods, and applications.* John Benjamins Publishing Company.

Casas Gómez, M. 2022. Docuterm: Modelo de documentación de usos terminológicos. *Revista de Lingüística y Lenguas Aplicadas (RLLA)*, 17:11–27.

Chun, Y., M. Kim, D. Kim, C. Park, and H.-S. Lim. 2025. Enhancing automatic term extraction with large language models via syntactic retrieval. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 9916–9926.

Corpas Pastor, G. and M. Seghiri Domínguez. 2010. Size matters: A quantitative approach to corpus representativeness. In *Lengua, traducción, recepción.* Publicaciones Universidad de León.

da Silva Conrado, M., A. D. Felippo, T. A. Salgueiro Pardo, and S. O. Rezende. 2014. A survey of automatic term extraction for brazilian portuguese. *Journal of the Brazilian Computer Society*, 20(1):12.

Daille, B. 2003. Conceptual structuring through term variations. In *Proceedings of the ACL 2003 workshop on Multiword expressions: analysis, acquisition and treatment*, pages 9–16.

Durán Muñoz, I. 2011. *El trabajo ontoterminográfico aplicado a la traducción de textos del turismo de aventura (español, inglés, alemán): fases de preparación.* Ph.D. thesis, Universidad de Málaga, Servicio de Publicaciones.

Estopà, R. 1999. Eficiencia en la extracción automática de terminología. *Perspectives: Studies in Translatology*, 7(2):277–286.

Fkih, F., M. N. Omri, and I. Toumia. 2012. A linguistic model for terminology extraction based conditional random fields. *ArXiv*, abs/1210.0252.

Frantzi, K., S. Ananiadou, and H. Mima. 2000. Automatic recognition of multiword terms:. the c-value/nc-value method. *International journal on digital libraries*, 3(2):115–130.

Heylen, K. and D. De Hertog. 2015. Automatic term extraction. In *Handbook of terminology.* John Benjamins Publishing Company, pages 203–221.

Jacquemin, C. and D. Bourigault. 2005. Term extraction and automatic indexing. In R. Mitkov, editor, *The Oxford Handbook of Computational Linguistics.*

Kageura, K. and B. Umino. 1996. Methods of automatic term recognition: A review. *Terminology. International Journal of Theoretical and Applied Issues in Specialized Communication*, 3(2):259–289.

Kockaert, H. J., F. Steurs, A. Alsulaiman, and A. Allaithy. 2015. *Handbook of terminology*, volume 1. John Benjamins Publishing Company Amsterdam.

Koza Orellana, W. 2015. Proposal for automatic extraction of medical term candidates with linguistic information processing description and evaluation of results. *Alfa: Revista de Linguística (São José do Rio Preto)*, 59:113–128.

Kÿto, M., J. Rudanko, and E. Smitterberg. 2000. Building a bridge between the present and the past: A corpus of 19th-century english. *ICAME journal*, 24:85–98.

Lafferty, J., A. McCallum, and F. C. Pereira. 2001. Conditional random fields: Probabilistic models for segmenting and labeling sequence data.

Levy, O., S. Remus, C. Biemann, and I. Dagan. 2015. Do supervised distributional methods really learn lexical inference relations? In *Proceedings of the 2015 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 970–976.

L'Homme, M.-C. 2020. *Lexical Semantics for Terminology*. John Benjamins.

Lossio-Ventura, J. A., C. Jonquet, M. Roche, and M. Teisseire. 2016. Biomedical term extraction: overview and a new methodology. *Information Retrieval Journal*, 19:59–99.

Martín Chozas, P. and P. Calleja Ibáñez. 2018. Challenges of terminology extraction from legal spanish corpora. CEUR Workshop Proceedings.

Nazar, R. 2016. Distributional analysis applied to terminology extraction: First results in the domain of psychiatry in spanish. *Terminology. International Journal of Theoretical and Applied Issues in Specialized Communication*, 22(2):141–170.

Peng, F., F. Feng, and A. McCallum. 2004. Chinese segmentation and new word detection using conditional random fields. In *Proceedings of the 20th International Conference on Computational Linguistics*, page 562. Association for Computational Linguistics.

Rigouts Terryn, A., V. Hoste, and E. Lefever. 2022. Tagging terms in text: A supervised sequential labelling approach to automatic term extraction. *Terminology*, 28(1):157–189.

Stanković, R., C. Krstev, I. Obradović, B. Lazić, and A. Trtovac. 2016. Rule-based automatic multi-word term extraction and lemmatization. In *LREC*, pages 507–514.

Terryn, A. R., V. Hoste, P. Drouin, and E. Lefever. 2020. Termeval 2020: Shared task on automatic term extraction using the annotated corpora for term extraction research (acter) dataset. In *Proceedings of the 6th International Workshop on Computational Terminology*, pages 85–94.

Tran, H. T. H., C.-E. González-Gallardo, J. Delaunay, A. Doucet, and S. Pollak. 2024. Is prompting what term extraction needs? In *International Conference on Text, Speech, and Dialogue*, pages 17–29. Springer.

Tran, H. T. H., M. Martinc, J. Caporusso, A. Doucet, and S. Pollak. 2023. The recent advances in automatic term extraction: A survey. *arXiv preprint arXiv:2301.06767*.

Vivaldi, J. and H. Rodríguez. 2011. Extracting terminology from wikipedia. *Procesamiento del lenguaje natural*, 47:65–73.

Zhang, Q., Q. Lu, and Z. Sui. 2007. Measuring termhood in automatic terminology extraction. In *2007 International Conference on Natural Language Processing and Knowledge Engineering*, pages 328–335.

Zhang, X., Y. Song, and A. C. Fang. 2010. Term recognition using conditional random fields. In *Proceedings of the 6th International Conference on Natural Language Processing and Knowledge Engineering*, pages 1–6.

Zhang, Y., H. Zhou, and Z. Li. 2020. Fast and accurate neural crf constituency parsing. In C. Bessiere, editor, *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence, IJCAI-20*, pages 4046–4053. International Joint Conferences on Artificial Intelligence Organization, 7. Main track.

Zheng, D., T. Zhao, and J. Yang. 2009. Research on domain term extraction based on conditional random fields. In W. Li and D. Mollá-Aliod, editors, *Computer Processing of Oriental Languages. Language Technology for the Knowledge-based Economy*, volume 5459 of *Lecture Notes in Computer Science*. Springer, Berlin, Heidelberg, pages 237–248.